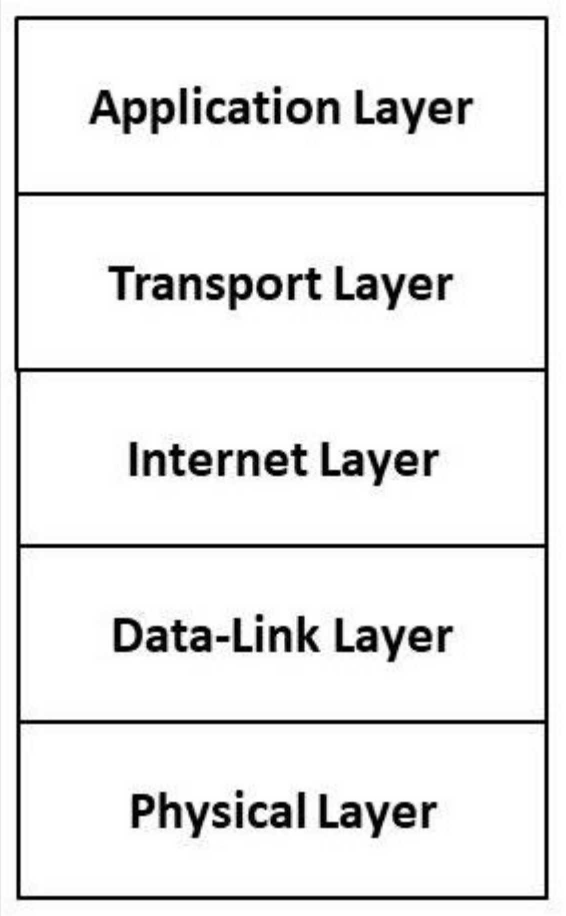
COMMUNICATION SOFTWARE

Presented by:
Goutham G
Vignesh B

Communication Software

TCP/IP Protocol Suite

- It was developed during the initial days of research on the internet.
- This research led to the most important concept of **Packet Switching**.



Layers of TCP/IP Protocol Suite

- Physical layer
- Data link layer (referred also as Network layer)
- Internet Protocol (IP) layer
- Transport layer (TCP layer and UDP layer)
- Application layer

□ Physical layer:

- This layer defines the characteristics of the transmission such as **data rate** and signal **encoding scheme**.

□ Data Link layer:

- This layer defines the protocols to manage the links- **establishing a link**, **transferring the data received from the upper layers**, and **disconnecting the link**.
- In LAN, the data link layer id divided into two sub layers:

  Medium Access Control (MAC) sub layer

  Logical Link Control (LLC) sub layer

□ Internet Protocol (IP) layer:

- The two important functions of this layer are **addressing** and **routing**.
- IP layer functionality is implemented in software
- IP layer software runs on every end system and router is connected to the internet.

- The presently running IP layer software is called **IP version 4**.
- This version will be slowly replaced by **IP version 6**.
- Each system connected to the internet is given a unique address known as **IP address**.
- In **IP version 4**, the IP address length is **32 bits** and in **IP version 6**, it is **128 bits**.
- Most of the embedded systems use **32-bit** IP address.

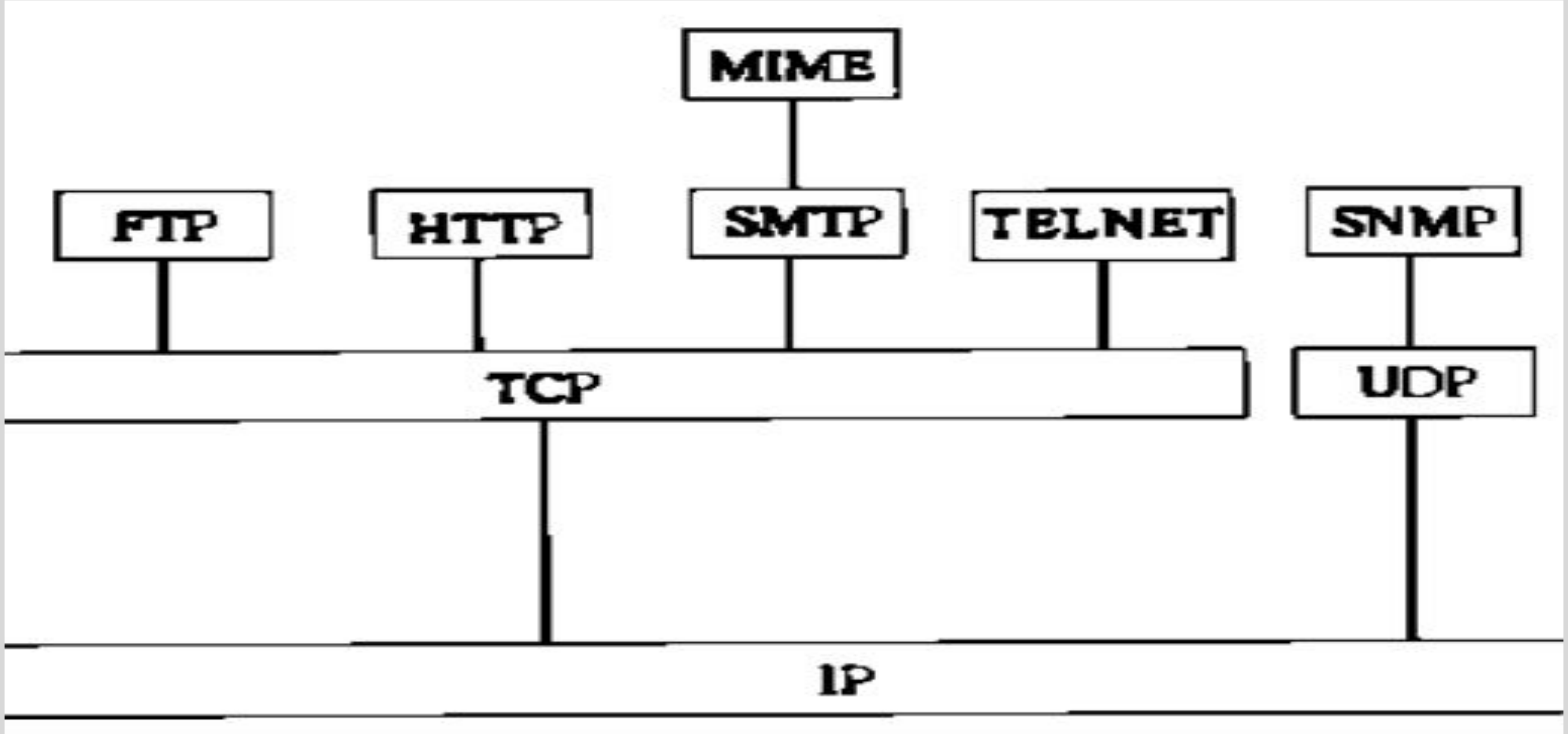
Transport layer:

- This layer provides **end-to end data transfer service** between **two systems connected to the internet**.
- Since **IP layer does not provide a reliable service**, it is the responsibility of the transport layer to incorporate reliability through acknowledgements , retransmissions, etc.
- The transport layer software **runs on every end system**.

Application layer:

- This layer differs from **application to application**.
- **Two processes** on **two end systems communicate** with application layer as interface.
- The application process generates an application byte stream which is divided into TCP segments and sent to the IP layer.
- The TCP segment is encapsulated in the IP datagram and sent to the data link layer.
- The IP diagram is encapsulated in the data link layer frame.
- The data link layer frame is sent to the physical layer interface and then the bit stream is sent over the transmission medium.

Application layer protocols in TCP/IP protocol stack



- Simple Mail Transfer Protocol (SMTP), for electronic mail containing ASCII text.
- Multimedia Internet Mail Extension (MIME), for electronic mail with multi-media content.
- File Transfer Protocol (FTP), for file transfer.
- TELNET for remote login.
- Hyper Text Transfer Protocol (HTTP), for World Wide Web service.
- Simple Network Management Protocol (SNMP), for network management. **Note** that SNMP runs above the UDP layer and not the TCP layer.

To make an embedded system network-enabled, the TCP/IP protocol stack is integrated with the operating system software and the application software. The entire software is the transferred to the memory of the embedded system.

Thank you